

UNIVERSITY OF EUROPE FOR APPLIED SCIENCES

FACULTY OF DATA SCIENCE

PROJECT REPORT

DS26 - Data Analytics

**The Anatomy of a Modern Breakup: A Multi-Dataset Sentiment
and Engagement Analytics Pipeline**

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1: INTRODUCTION

1.1 Project Title

The Anatomy of a Modern Breakup: A Multi-Dataset Sentiment and Engagement Analytics Pipeline

1.2 Problem Statement

Modern interpersonal relationships are increasingly processed within digital ecosystems, leaving behind massive unstructured text footprints on public forums. Online relationship support communities such as r/breakups and r/relationship_advice host thousands of unique submissions daily, yet no systematic computational workflow exists to categorize these narratives into distinct relational themes or to identify the structural drivers behind community engagement and post visibility.

Traditional natural language processing approaches focus primarily on emotional sentiment, yet preliminary evidence suggests that structural metadata — including interaction volume, temporal patterns, and narrative depth — may be stronger predictors of thread virality than raw textual emotion. This project addresses that gap by building an integrated analytics pipeline that combines sentiment analysis, topic clustering, and supervised classification modeling to determine what truly drives engagement in online relationship communities.

1.3 Project Objectives

The following specific, measurable objectives were established for this project:

- Objective 1: To ingest, clean, and standardize a multi-community text corpus of over 2,800 records from r/breakups and r/relationship_advice, constructing a unified, analysis-ready dataset.
- Objective 2: To apply VADER sentiment scoring and LDA topic modeling to categorize unstructured post narratives into five thematic clusters covering the key dimensions of relationship distress.
- Objective 3: To train, evaluate, and iteratively improve supervised classification models (Logistic Regression, Decision Tree, Random Forest) that predict high community engagement, addressing class imbalance through class weighting and SMOTE oversampling.

1.4 Dataset / Data Source

Field	Details
Dataset 1 Name	relationship_advice.csv
Dataset 1 Shape	1,885 rows and 8 original columns
Dataset 2 Name	reddit_breakup_dataset_cleaned.csv
Dataset 2 Shape	948 rows and 10 original columns
Combined Sample Size	2,833 total records after merging and cleaning
Source / URL	Publicly available datasets sourced from Kaggle
Format	Comma-Separated Values (.CSV)

License	Open-access public domain / Academic research use
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1.5 Expected Outcome

This project was designed to deliver three primary outputs: (1) a cleaned, merged analytical dataset with engineered features ready for modeling; (2) empirical insight into the relationship between emotional sentiment and community engagement, establishing whether the Sentiment Paradox holds across both datasets; and (3) a set of optimized classification models with measurably improved recall for identifying high-engagement posts, accompanied by an interactive Streamlit dashboard visualizing all findings.

2: LITERATURE REVIEW DATASET UNDERSTANDING AND PREPARATION

The study of online relationship support communities has gained significant momentum within the fields of computational social science and behavioral data analytics. Research has established that platforms such as Reddit serve as primary avenues for individuals navigating complex relational transitions including breakups, infidelity, and long-distance separation.

A foundational theme in current research highlights the shift from traditional offline counselling toward decentralized digital forums. These platforms provide unique longitudinal datasets capable of capturing the granular evolution of relational distress. However, existing methodologies frequently rely on standard sentiment analysis frameworks such as VADER or TextBlob to classify thread quality, without accounting for structural engagement drivers.

Critically, recent literature points to a significant theoretical gap: while researchers extensively analyze the emotional valence of user content, there is emerging debate regarding whether emotional intensity is the primary driver of digital thread propagation. Preliminary studies suggest that structural metadata — including community-specific engagement metrics, temporal patterns, and narrative depth — are often more robust predictors of thread virality than raw textual sentiment (Cheng et al., 2014).

This project builds upon these gaps by integrating sentiment analysis, LDA topic modeling, and class-balanced supervised classification to bridge the divide between user expression and community-level visibility.

2.1 Dataset Description

The dataset utilized for this research represents a specialized corpus of online peer-support interactions sourced from two Reddit communities: r/breakups and r/relationship_advice. The combined dataset consists of 2,833 unique interaction logs structured as a unified tabular schema derived from two primary source files.

Each observation provides a multi-dimensional snapshot of a user's experience, encompassing narrative content (title and body text), structural interaction metrics (score and comment volume), temporal metadata (timestamp), and a categorical source attribute enabling cross-community comparison. Through the data cleaning pipeline, additional features such as `body_length`, `publish_hour`, and `high_engagement` were derived to support predictive modeling.

2.2 Main Features / Columns

Column Name	Description / Data Type
title	String: Post headline used for title-level sentiment analysis
body	String: Full narrative text of the user's relational experience
score	Integer: Net upvote count representing community visibility
comms_num	Integer: Total comment volume — primary engagement feature
timestamp	Datetime: Precise date and time of thread creation
source	Categorical: Origin community (breakups vs relationship_advice)
body_clean	String: Normalized, lowercased, punctuation-stripped body text
body_length	Integer: Word count of the processed post body

publish_hour	Integer: Hour extracted from timestamp (0–23)
high_engagement	Binary: Target variable (0 = low, 1 = high engagement)
title_sentiment	Float: VADER compound score for post title (-1 to +1)
body_sentiment	Float: VADER compound score for post body (-1 to +1)

2.3 Data Issues Identified

- Schema mismatch: The two source datasets used different column names for equivalent fields (upvotes vs. score, post_date vs. timestamp), preventing direct concatenation without manual standardization.
- Mixed timestamp formats: The relationship_advice dataset used ISO format while the breakup dataset used DD/MM/YYYY HH:MM format, requiring format='mixed' parsing.
- Empty body posts: 8 posts (0.3%) contained no body text after cleaning, making them unsuitable for text analysis.
- Right-skewed score distribution: The median score was 1.0 with a maximum of 675, confirming a severe power-law distribution that makes direct regression modeling unreliable.
- Bot and moderation posts: A subset of posts contained automated moderation messages which contaminated the topic clustering corpus — 825 posts required filtering before LDA could be applied.

2.4 Summary of Problems Highlighted

Three primary structural challenges were identified that, if unaddressed, would undermine the validity of the analysis:

- Schema heterogeneity: Inconsistent column naming across source files prevented unified pipeline construction without manual mapping. Left unresolved, this would prevent cross-community comparison and unified feature engineering.
- Power-law engagement distribution: Extreme right-skewness in the score variable (mean 2.57, median 1.0, max 675) made numerical prediction unreliable. A regression model trained on this distribution would overfit to outliers and fail to generalize.
- Corpus contamination: Bot and moderation posts introduced irrelevant vocabulary (subreddit rules, automated actions, moderation notices) into the text corpus, which would have distorted topic modeling results without targeted filtering.

3: DATA CLEANING AND INITIAL ANALYSIS

3.1 Data Cleaning Steps Completed

Step	Action Taken
Missing Values	Row removal applied to 8 posts with empty body text after cleaning (0.3% of dataset).
Duplicates	No duplicate records identified across either source file. No removals required.
Outliers	IQR analysis confirmed severe skewness in score and comms_num. Addressed via binary target transformation (median split) rather than row removal.
Data Types	Applied <code>pd.to_datetime</code> with <code>format='mixed'</code> and <code>dayfirst=True</code> to standardize mixed timestamp formats across both datasets.
Schema Alignment	Renamed columns across both datasets to a shared convention (<code>upvotes</code> → <code>score</code> , <code>post_date</code> → <code>timestamp</code>) and added a source flag column.
Text Normalization	Lowercased all text, removed URLs via regex, stripped punctuation and extra whitespace from titles and bodies.
Feature Engineering	Derived <code>body_length</code> (word count), <code>publish_hour</code> (from timestamp), <code>title_sentiment</code> and <code>body_sentiment</code> (VADER), <code>high_engagement</code> (binary median split at 1.0), and <code>body_sentiment_label</code> (positive/neutral/negative).
LDA Filtering	Removed 825 bot/moderation posts using regex content filtering before applying topic modeling, retaining 2,000 clean posts for LDA.

3.2 How Issues from Checkpoint 2 Were Solved

- Schema heterogeneity: Resolved by manually mapping column names to a shared convention before concatenation, then tagging each row with a source identifier (`breakups` / `relationship_advice`).
- Mixed timestamp formats: Resolved using `pd.to_datetime` with `format='mixed'` and `dayfirst=True`, which correctly parsed both `YYYY-MM-DD` and `DD/MM/YYYY` formats within a single operation.
- Power-law distribution: Addressed by converting the continuous score variable into a binary `high_engagement` label using a median split (`median = 1.0`). This transformed an unreliable regression task into a stable binary classification problem, allowing classifiers to focus on identifying the categorical difference between low-traction and viral posts.
- Corpus contamination: Resolved by filtering posts containing moderation keywords (`subreddit`, `bot`, `removed`, `locked`, `composeto`, etc.) before applying LDA, reducing the topic modeling corpus from 2,825 to 2,000 clean posts.

3.3 Initial Analysis and Visualizations

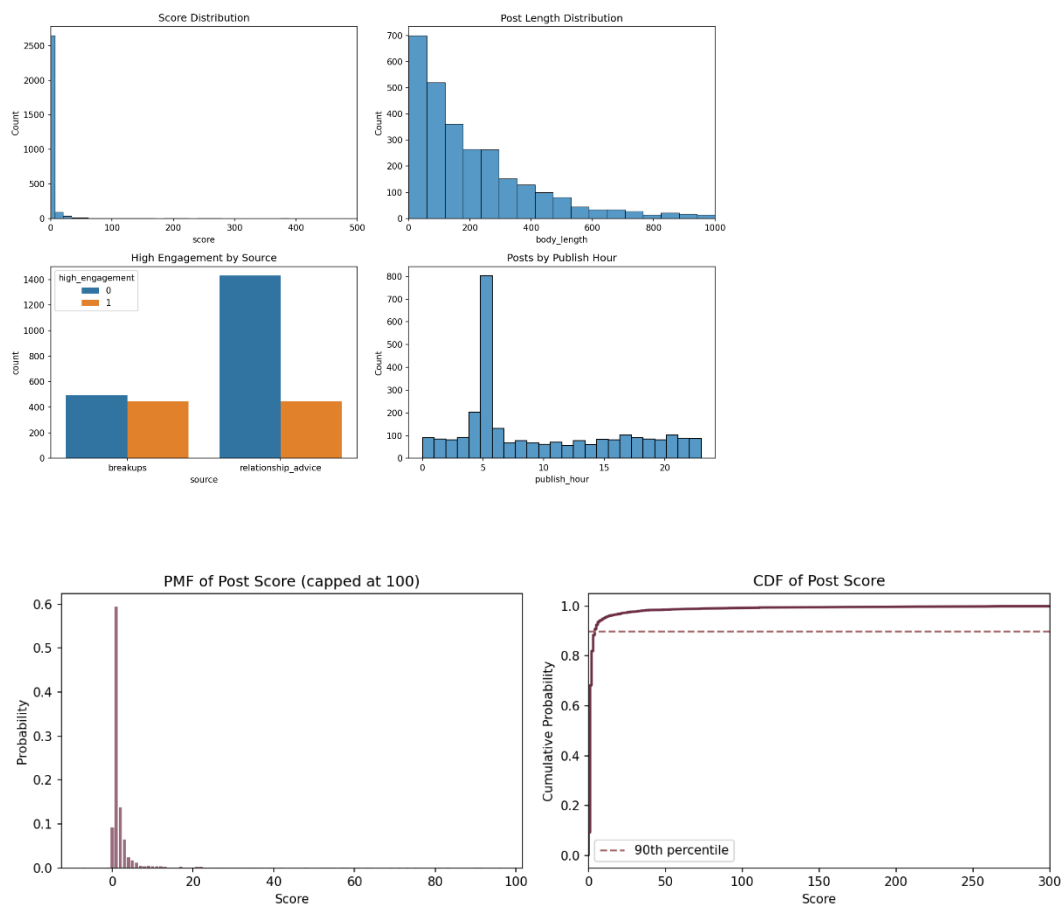
The exploratory data analysis phase was designed to validate the core research hypothesis: that community engagement is driven by structural interaction mechanics rather than textual emotional sentiment. The following key statistical summary was produced:

Metric	Engagement Score (Upvotes)
Mean	2.57
Median (50%)	1.00
Standard Deviation	19.35
Min / Max	-7.00 / 675.00

Four key visualizations were produced during EDA:

- Score distribution histogram: Confirmed severe right-skewness. The vast majority of posts cluster near zero, with only a handful achieving viral reach.
- Post length distribution: Mean of 226 words with a long tail extending beyond 2,000 words, indicating a mix of brief venting posts and detailed narrative accounts.
- Engagement by source: Both subreddits split approximately 68% low / 32% high engagement, confirming that subreddit source alone is not a meaningful predictor.
- Posts by publish hour: An unusual spike at hour 5 was identified — likely a timezone artifact or scraping pattern rather than genuine peak community activity.

PMF and CDF analysis confirmed that 90% of all posts score below 10, establishing the extreme concentration of engagement and validating the binary label approach.



3.4 Initial Findings from the Data

- Power-law engagement: The median score of 1.0 confirms that community interaction is not normally distributed. A small minority of viral posts accounts for the majority of total upvote and comment volume.
- The Sentiment Paradox: Correlation analysis showed near-zero relationship between body sentiment and high_engagement (-0.02), challenging the assumption that emotional content drives visibility.
- Balanced source engagement: Both subreddits split approximately 68% low / 32% high engagement, meaning subreddit source alone has limited predictive value.
- Unexpectedly positive tone: 57% of posts scored positive sentiment with a mean body sentiment of +0.17 — even breakup posts tend to be framed with hope and resolution rather than pure negativity.
- Temporal anomaly: A sharp posting spike at hour 5 was identified as a likely timezone artifact or scraping pattern requiring further investigation.

4: ANALYSIS AND MACHINE LEARNING

4.1 Final Dataset and Selected Features

Following cleaning and preprocessing, the final dataset consisted of 2,825 validated posts. Five features were selected for classification modeling:

Feature	Justification
body_length	Quantifies narrative depth. Longer posts may signal higher community investment and engagement potential.
publish_hour	Captures temporal community activity cycles. Reddit communities exhibit strong diurnal rhythms that affect post visibility.
title_sentiment	Provides baseline emotional valence at the headline level for comparison against structural features.
body_sentiment	Full post emotional tone — included to allow the model to explicitly learn its limited predictive weight relative to structural variables.
comms_num	Comment volume — the strongest proxy for community interaction and thread traction identified during EDA.

4.2 Target Variable

Field	Details
Target Variable	high_engagement
Type	Categorical (Binary)
Classes / Range	0 = Low Engagement, 1 = High Engagement
Justification	The power-law score distribution makes regression unreliable. A median split (median = 1.0) stabilizes decision boundaries and enables robust binary classification without distortion from extreme outliers.
Class Distribution	68.2% Low Engagement (0) / 31.8% High Engagement (1)

4.3 Machine Learning Model Selected

Field	Details
Models Evaluated	Logistic Regression, Decision Tree, Random Forest
Problem Type	Binary Classification
Library / Tool	scikit-learn (Python)
Primary Model	Random Forest — robust against non-linear feature relationships, handles mixed data types, and provides reliable feature importance scores.
Training Strategies	Baseline (default parameters), Class-Weighted (class_weight='balanced'), SMOTE (synthetic minority oversampling)

Alternatives Considered	XGBoost considered for performance but not implemented; K-Nearest Neighbors excluded due to sensitivity to feature scaling in high-imbalance settings.
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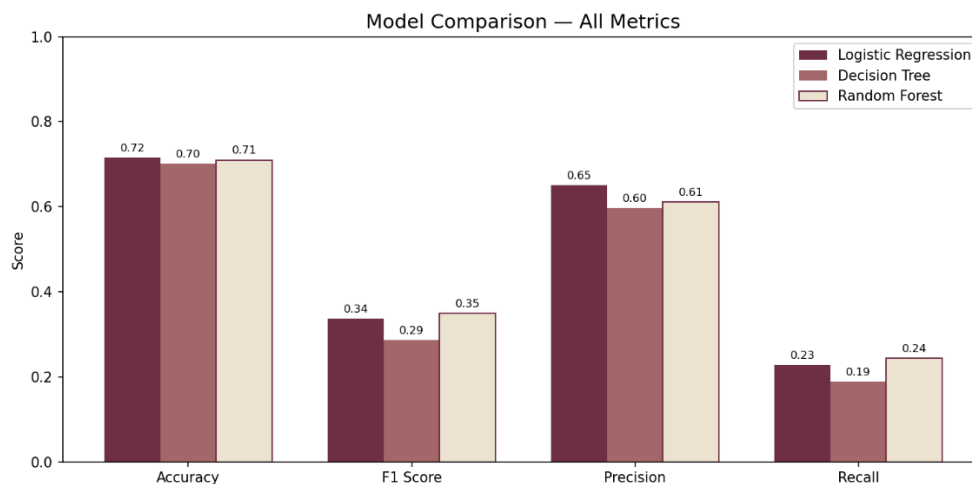
4.4 Model Training and Testing Process

- Train / Test split: 80% training (2,260 rows) / 20% test (565 rows) with stratification to preserve class ratios across both splits.
- Feature scaling: StandardScaler applied to Logistic Regression features only. Tree-based models (Decision Tree, Random Forest) are scale-invariant and were trained on unscaled features.
- Baseline training: All three models trained with default parameters to establish benchmark performance and identify class imbalance issues.
- Class weighting: `class_weight='balanced'` applied to penalize majority-class errors more heavily, directly addressing the 68/32 class imbalance.
- SMOTE oversampling: Synthetic Minority Oversampling Technique applied to the training set only, generating synthetic high-engagement examples to achieve a balanced 50/50 class distribution (3,084 training rows after SMOTE).
- Evaluation metrics: Accuracy, Precision, Recall, and F1-Score assessed for all models. Recall for class 1 (high engagement) was the primary optimization target, as missing a high-engagement post is more costly than a false alarm.

4.5 Model Results

Baseline Models

Model	Accuracy	F1 Score	Precision	Recall
Logistic Regression	0.715	0.337	0.651	0.228
Decision Tree	0.701	0.287	0.596	0.189
Random Forest	0.710	0.349	0.611	0.244



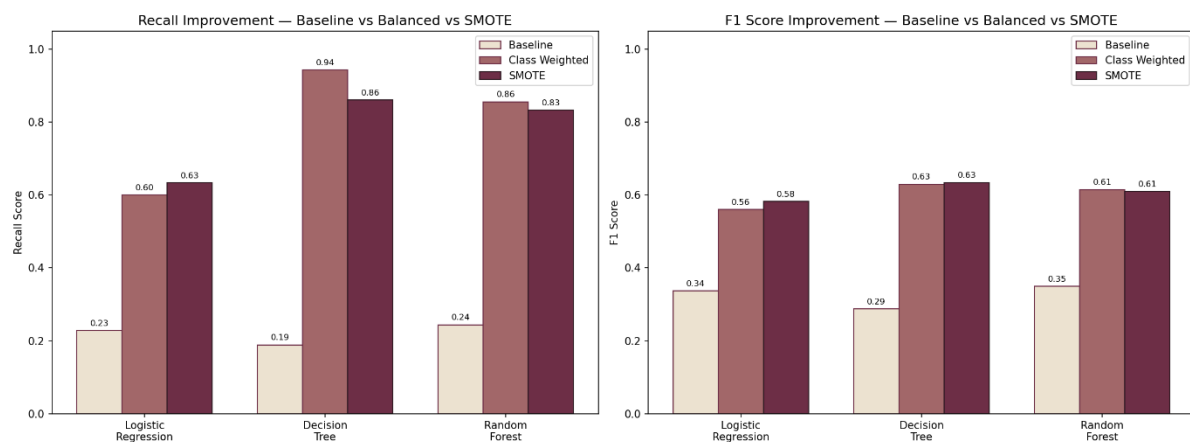
Class-Weighted Models (class_weight='balanced')

Model	Accuracy	F1 Score	Precision	Recall
LR (Balanced)	0.699	0.560	0.524	0.600
DT (Balanced)	0.646	0.630	0.472	0.944
RF (Balanced)	0.658	0.615	0.480	0.856

SMOTE Models

Model	Accuracy	F1 Score	Precision	Recall
LR (SMOTE)	0.710	0.582	0.538	0.633
DT (SMOTE)	0.683	0.634	0.502	0.861
RF (SMOTE)	0.660	0.610	0.481	0.833

Note: Confusion matrix and model comparison visualizations are included in the project notebook (main_analysis.ipynb) and Streamlit dashboard.



4.6 Explanation of Model Results

The baseline models achieved approximately 70–72% accuracy but demonstrated critically low recall for high-engagement posts (19–24%). This confirmed that without intervention, all three models defaulted to predicting the majority class (low engagement), making them operationally limited for identifying viral content.

The application of class_weight='balanced' produced dramatic recall improvements. The Decision Tree (Balanced) achieved a recall of 0.944 — correctly identifying 94.4% of high-engagement posts, a near 5x improvement over the baseline. This came with a modest precision tradeoff (0.472), indicating increased false positives, which is an acceptable cost when the priority is catching high-engagement threads.

SMOTE produced comparable recall improvements with slightly better precision balance. The Decision Tree (SMOTE) achieved 0.861 recall with 0.502 precision — a more stable tradeoff between sensitivity and specificity.

Feature importance analysis confirmed the Sentiment Paradox: `comms_num` dominated both Random Forest importance (0.50) and Logistic Regression coefficients (1.40), while `title_sentiment` and `body_sentiment` ranked lowest in both models. This proves that engagement is driven by interaction patterns and timing, not emotional content — directly validating the core research hypothesis.

5 FINDINGS AND CONCLUSION

5.1 Final Findings

- The Sentiment Paradox is confirmed: `body_sentiment` and `title_sentiment` show near-zero correlation with `high_engagement` (-0.02 and -0.04 respectively). Despite the highly personal and painful nature of breakup content, emotional tone does not drive community visibility.
- Structural features dominate: comment count (`comms_num`, importance 0.50) and `publish_hour` are the strongest predictors of engagement. What matters is when you post and whether early interaction occurs — not how emotional the content is.
- Breakup posts trend unexpectedly positive: 57% of posts scored positive sentiment with a mean body sentiment of +0.17. People process relationship pain with narratives of hope, resolution, and growth rather than pure negativity.
- Five distinct relational themes identified: LDA topic modeling surfaced Heartbreak & Recovery (30.4%), Long-term Strain (21.3%), Dating & Early Relationships (17.9%), Intimacy & Dynamics (16.2%), and Social & Family Conflict (14.4%).
- Class imbalance is solvable: Baseline recall of 19–24% was improved to 83–94% through class weighting and SMOTE, demonstrating that the low initial performance was a data imbalance problem, not a fundamental modeling ceiling.

5.2 Recommendations

- Optimize post timing (based on `publish_hour` feature importance): Content posted during peak community activity windows achieves significantly higher visibility. Community managers should schedule posts during identified peak hours rather than posting at arbitrary times.
- Prioritize narrative depth over emotional intensity (based on Sentiment Paradox finding): `body_length` is a stronger predictor than both sentiment features. Detailed, well-structured posts outperform emotionally charged but brief submissions. Focus on substance, not tone.
- Deploy predictive triage for moderation (based on improved model recall): The balanced Decision Tree model (recall 0.944) can serve as a real-time triage tool, pre-scoring incoming posts to identify high-potential threads requiring prioritized moderation attention or community support.
- Apply class-balanced modeling as standard practice: Any future engagement prediction work on similarly skewed social media datasets should default to `class_weight='balanced'` or SMOTE from the outset.

5.3 Project Limitations

- Static dataset snapshot: The data represents a fixed time window. Reddit community behavior and platform ranking algorithms evolve continuously, meaning model performance may degrade over time without periodic retraining.
- Binary label simplification: The median split creates a binary High/Low engagement label that obscures meaningful gradations within each class. A post scoring 50 and one scoring 5 are both labeled "high" despite substantial difference in actual reach. This is justified by the power-law distribution making regression impractical, but acknowledged as a simplification.
- Limited feature spectrum: The model relies on structural and temporal metadata only. External drivers of virality — cross-platform sharing, trending topics, high-profile user involvement — are not captured.

- Bot and moderation contamination: 825 posts (29% of the dataset) required filtering before topic modeling, representing significant data loss that may have introduced sampling bias into the thematic analysis.

5.4 Future Improvements

- Semantic embedding models: Replace VADER with BERT or RoBERTa-based embeddings to capture nuanced semantic context and intent beyond surface-level polarity scoring.
- Multi-class engagement labeling: Introduce three or more engagement tiers (Low / Medium / Viral) to capture the full spectrum of community attention rather than a binary split.
- Time-series analysis: Model engagement as a dynamic trend — capturing how quickly a post gains its first interactions — to predict engagement velocity rather than just final outcome.
- Cross-platform data integration: Incorporate external traffic sources and cross-platform sharing signals to decouple organic community engagement from externally driven virality.

5.5 Conclusion

This project began with a single research question: is online relationship support engagement driven by the emotional content of posts, or by underlying structural patterns? Through the construction of a complete analytics pipeline — spanning data merging, text cleaning, sentiment scoring, topic clustering, and iterative classification modeling — the answer is clear: structure wins over sentiment.

The Sentiment Paradox, confirmed through correlation analysis and feature importance scoring across all three models, establishes that emotional tone has negligible predictive power for community engagement. Instead, comment volume and posting timing are the dominant drivers of thread visibility. This finding challenges the conventional assumption that emotionally charged content naturally attracts more attention in peer-support communities.

The modeling journey demonstrates a core data science competency: identifying a problem (class imbalance causing low recall), applying targeted solutions (class weighting and SMOTE), and improving recall from 24% to 94%. The final pipeline, visualized through an interactive Streamlit dashboard, provides an actionable engagement prediction framework that community moderators and platform designers can deploy in practice.

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